



AALBORG UNIVERSITY  
DENMARK

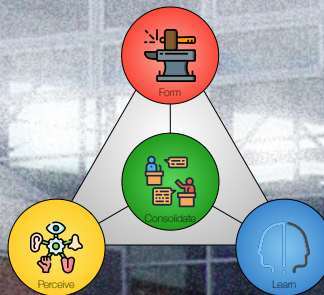
# 9 EVOLVABILITY & GENERAL INNOVATION THEORY

ESSENCE – CHAPTER 13

ROSE, J. (2010) SOFTWARE INNOVATION: EIGHT WORK-  
STYLE HEURISTICS FOR CREATIVE SOFTWARE  
DEVELOPERS

IVAN AAEN  
INSTITUT FOR DATALOGI  
[IVAN@CS.AAU.DK](mailto:IVAN@CS.AAU.DK)

SWI 2026

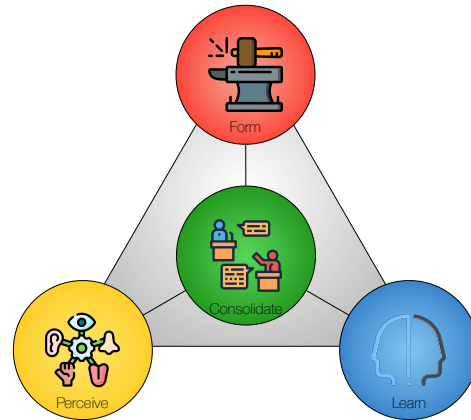


# Change of Course Plan – Lectures 9-11

| Module | Date | 8.15 - 10.00                                                                                                  | 10.15 - 12.00                                                                                       |
|--------|------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 1      | 4/2  | <b>Exercise 0 (Innovation opportunities)</b><br>(Innovation opportunities for you in this semester)           | <b>Lecture 1 (Introduction, Roles and Core Activities)</b><br>(Essence Ch. 1 & 2)                   |
| 2      | 11/2 | <b>Exercise 1 (Innovation in your project)</b><br>(Type, what, why, how, complexities, uncertainties)         | <b>Lecture 2 (The Problem-Solution Canvas)</b><br>(Essence Ch. 3 & 4)                               |
| 3      | 18/2 | <b>Exercise 2 (Start on PSC)</b><br>(Choose Fulcrum and start on PSC)                                         | <b>Lecture 3 (Rationale &amp; Problem)</b><br>(Essence Ch. 5 & 6)                                   |
| 4      | 25/2 | <b>Exercise 3 (Problem Scenarios)</b><br>(Develop Problem Scenarios)                                          | <b>Lecture 4 (Leverage)</b><br>(Essence Ch. 7)                                                      |
| 5      | 4/3  | <b>Exercise 4 (Leverage Scenarios)</b><br>(Develop Leverage Scenarios)                                        | <b>Lecture 5 (Solution)</b><br>(Essence Ch. 8)                                                      |
| 6      | 11/3 | <b>Exercise 5 (Solution Scenarios)</b><br>(Develop Solution Scenarios)                                        | <b>Lecture 6 (Horizon)</b><br>(Essence Ch. 9)                                                       |
| 7      | 18/3 | <b>Exercise 6 (Horizon)</b><br>(Develop Horizon and Horizon hypotheses)                                       | <b>Lecture 7 (Strategy &amp; Outer Environment)</b><br>(Essence Ch. 10 & 11)                        |
| 8      | 25/3 | <b>Exercise 7 (Outer Environment)</b><br>(Clarify Outer Environment)                                          | <b>Lecture 8 (Inner Environment)</b><br>(Essence Ch. 12)                                            |
| 9      | 8/4  | <b>Exercise 8 (Inner Environment)</b><br>(Clarify Inner Environment)                                          | <b>Lecture 9 (Evolvability &amp; General Innovation Theory)</b><br>(Essence Ch. 13 & Rose)          |
| 10     | 15/4 | <b>Exercise 9 (Evolvability)</b><br>(Diffusibility and adoptability)                                          | <b>Lecture 10 (Tactics, Manifestations, and Capabilities)</b><br><b>Guest Lecture</b> (Slides only) |
| 11     | 22/4 | <b>Exercise 10 (Tactics, Manifestations, and Capabilities)</b><br>(Tactics, Manifestations, and Capabilities) | <b>Lecture 11 (Merit)</b><br>(Essence Ch. 18)                                                       |
| 12     | 29/4 | <b>Exercise 11 (Merit)</b><br>(Merit and dependencies (ETVX))                                                 | <b>Lecture 12 (Innovation as Design Science &amp; Course Overview)</b><br>(Hevner & Gregor 2022)    |
| 13     | 6/5  | Work-module: Finish Mini-project                                                                              |                                                                                                     |

# Agenda

- Exercise 8 Follow-up.
- Chapter 13: Evolvability.
  - Diffusibility.
  - Adoptability.
  - Team Archetypes for Problem-solving and Innovation.
- General Innovation Theory
- Next time:
  - Exercise 9: Evolvability & General Innovation Theory.
  - Lecture 10: Tactics, Manifestations, Capabilities.
  - Guest lecture: Klaus Bundvig, Netcompany.



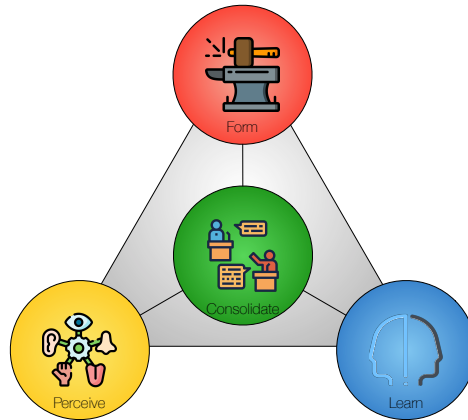
# Follow-up: Exercise 8



# Exercise 8 (Inner Environment)

This exercise is about **near-decomposability** and **VUCA** strategies for the **Inner Environment**.

- The introduction part of chapter 12 (pages 140-141) discusses the use of **conceptual models** for developing and revising design ideas. Which conceptual models do you use to design the architecture and outline primary components in your project?
- Are any of these models **helpful** for modeling near-decomposable designs, as discussed in Section 12.2?
- Discuss VUCA **strategies** for your Inner Environment (see Table 12.1). Are any of them relevant to your project? How? Suggest **alternative** strategies if they are not useful.
- How could you use these strategies **actively** in your project?



# Strategy

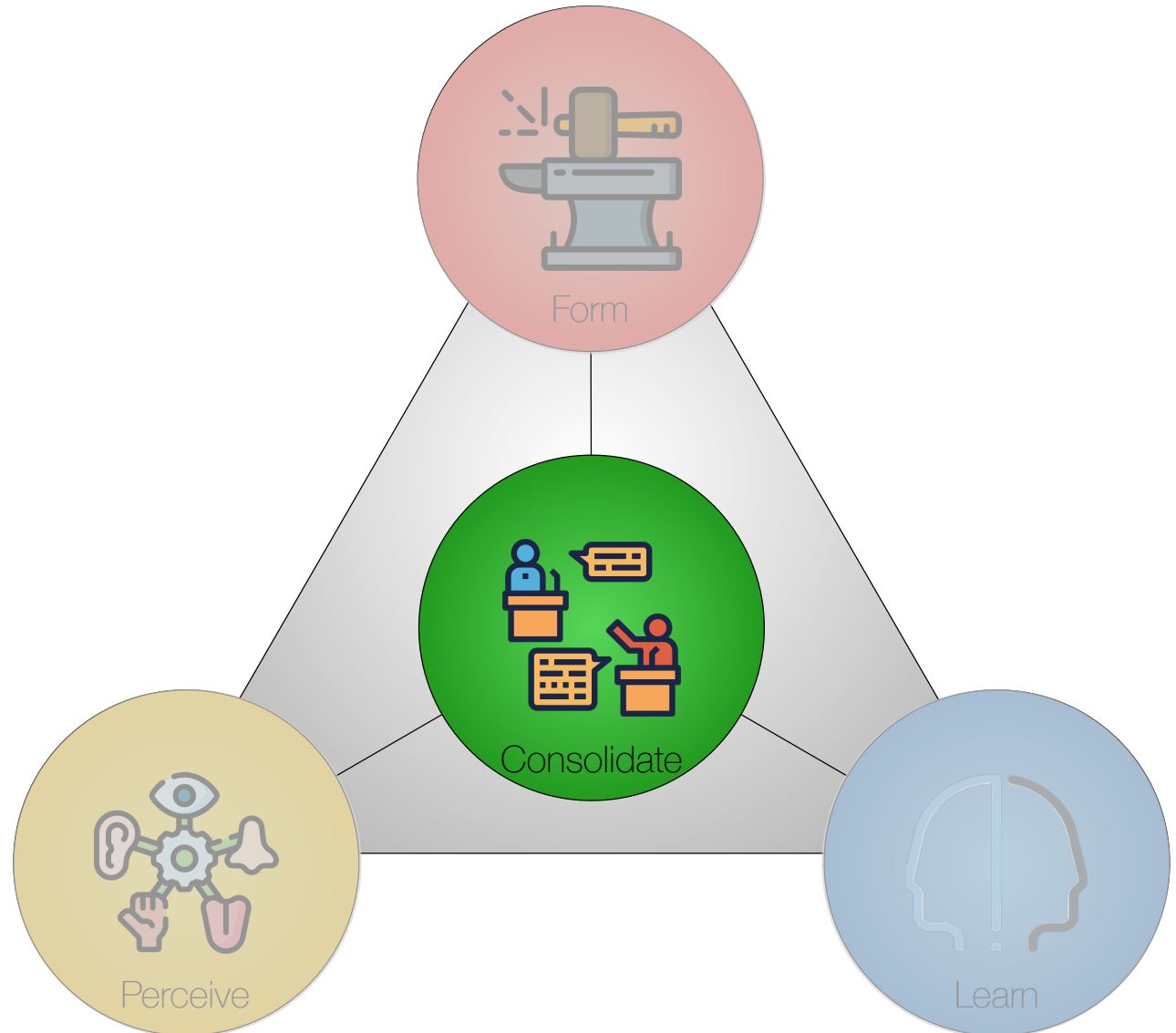


In preparing for battle, I have always found that plans are useless, but planning is indispensable.

Dwight D. Eisenhower (1890–1969)

# Four Core Activities

- *Perceive* – understanding the problem and its context.
- *Form* – designing the constructive parts of a solution.
- *Consolidate* – integrating the design contributions into a clear whole that addresses the situation.
- *Learn* – appraising the design.



# Three Levels of Abstraction



*Rationale* – the logical basis that makes actions meaningful and helps reason about strategy and tactics for solving the overall problem.

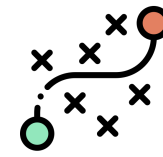


*Strategy* – the **overall plan** for addressing the problem, including its **scope** and **context**, the essential parts needed to build the solution, and any limitations or constraints that might influence its feasibility. At the strategy level, we also evaluate the potential for future **growth**.



*Tactics* – actions planned in accordance with the strategy to achieve specific ends.

# Consolidate Terminology



## Solution



Why the problem is resolved, why it is important, and why the result is appealing.

## Evolvability



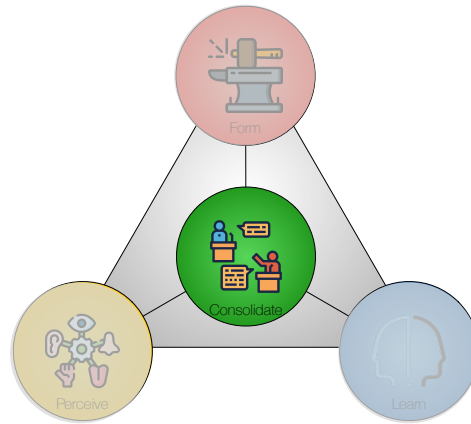
The capacity to adapt to outer environments or to add new leverage points to the inner environment.

## Merit

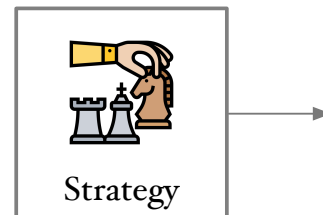
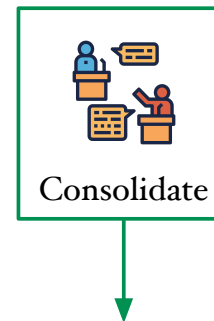


The value offered by the capabilities. The limitations of the solution and why these limitations are acceptable.





# Evolvability



*The advancement and diffusion of knowledge  
is the only guardian of true liberty*  
James Madison (1751–1836)

# Diffusion

**Diffusion** is crucial for economic growth. If a brilliant solution doesn't **spread**, it won't make a big **difference**.

- The value of an innovation depends on its wide **adoption**.
- We assume this also applies to the solutions created in a **specific project**.
- The term **evolvability** refers to the ability to successfully expand into other markets or contexts.
- Information and data exist in any social system, and we expect our technologies to have potential **across** various **domains**.
- **Diffusion** involves two steps: **moving** to different environments and **establishing** oneself there.

# Expanding the meaning of environments

- The **generic outer environment** refers to the context we design for in the current project, as well as similar or anticipated environments that may not yet exist.
- The **generic inner environment** builds on what we design in the current project, including optional leverage points to adapt it to changing demands or requirements.

# The Spider Plant Metaphor



# Diffusibility

We use the term **diffusibility** to describe the first part of **evolvability**: entering new environments—migrating across niches—and generating potential **variations** in search of new leverage points and application domains.

- **Variation** — creating probes (runners)
- **Selection** — choosing leverage points and domains that could become future bets on horizon H3, and using every opportunity to specialize probes toward those future bets.
- **Retention** — keeping strategic probes alive long enough to mature and demonstrate their value.

# Diffusibility and Fulcrums

- *Manifestation fulcrums*—if similar manifestations appear in many other contexts, there is potential for wider diffusion.
- *Capability fulcrums*—if similar capabilities support many different tasks, they could have broader use.
- *Merit fulcrums*—innovative ways to generate value and drive growth if we create value for users willing to pay for it.
- *Mission fulcrums*—overcoming new challenges or surpassing existing limits. Whether this *invention* becomes an *innovation* depends on its broader use and its impact.

# Adoptability

We use the term **adoptability** to describe the ability to **adapt to** and **fit into** new conditions. Adoptability refers to the **likelihood** that an innovation can be integrated into new domains.

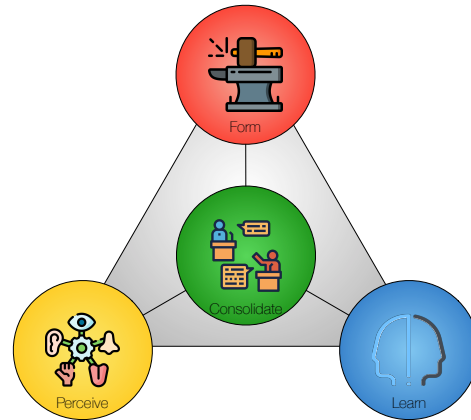
- Adoptability can be **enhanced by design**: modular solutions enable partial adoption or adaptation into other application areas.
- **Retention** leads to the mutual selection of **leverage points** and **application domains** that complement each other.

# Team Archetypes

|               |      | ADOPTABILITY                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                          |
|---------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DIFFUSIBILITY | HIGH | <b>HIGH ROLLERS</b><br>Many probes are used, but with a low survival rate. Very experimental but lacking clear direction. <ul style="list-style-type: none"> <li>• Focus on trying out options</li> <li>• Losses occur more often than gains</li> </ul>              | <b>PATHFINDERS</b><br>Able to navigate in unknown territories. Good at probing and good at landing or scaling. <ul style="list-style-type: none"> <li>• Focus on transformation</li> <li>• Optimal stance for unpredictable times</li> </ul>                             |
|               | LOW  | <b>RECREATORS</b><br>Neither exploring new options nor demonstrating the ability to create new leverage points or markets. <ul style="list-style-type: none"> <li>• Focus on conserving strengths</li> <li>• High-risk posture during times of disruption</li> </ul> | <b>STABLE PERFORMERS</b><br>Able to deliver well if the direction is known. Less suited for exploring H <sub>3</sub> options. <ul style="list-style-type: none"> <li>• Focus on delivering</li> <li>• Sensitive to disruptions in markets and leverage points</li> </ul> |
|               |      | Low                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                          |
|               |      |                                                                                                                                                                                                                                                                      | HIGH                                                                                                                                                                                                                                                                     |

*Table 13.1: Team Archetypes for Problem-solving and Innovation*





# General Innovation Theory

Rose, J. (2010) *Software Innovation: Eight work-style heuristics for creative software developers*, Aalborg, Software Innovation, Dept. of Computer Science, Aalborg University, pages referenced in slides.

Tidd, J., Bessant, J. R., and Pavitt, K. 2005 *Managing innovation: integrating technological, market and organization change*. Wiley.  
**Not part of syllabus**

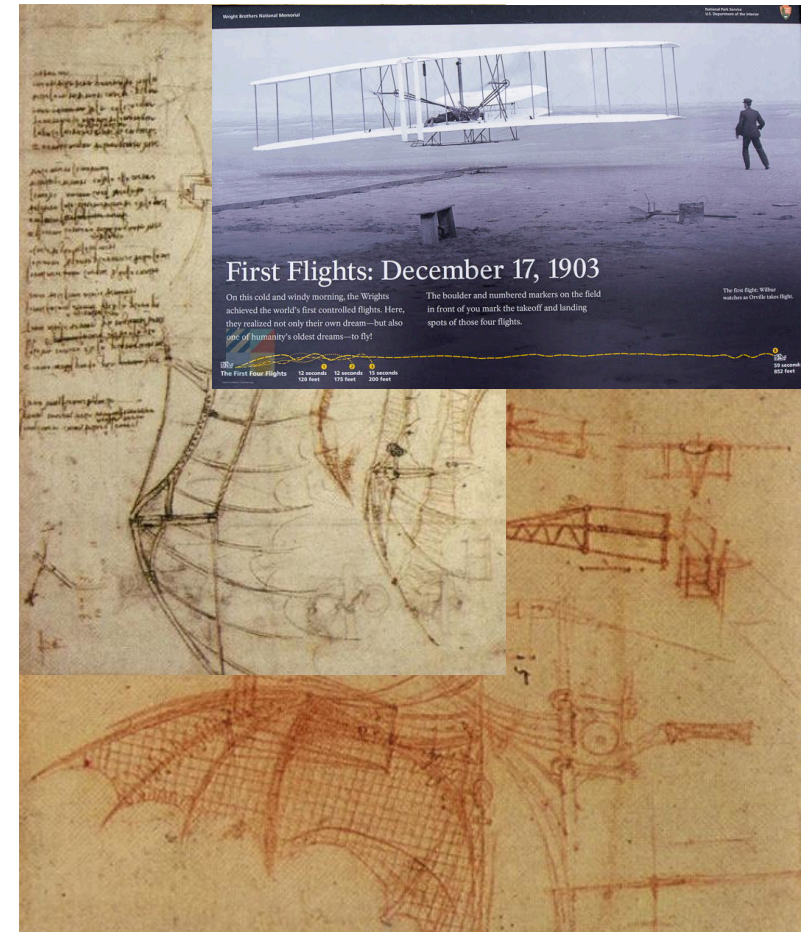
# Basic Terms

- *Creativity* – a state of mind which leads to inventive or innovative thinking.
- *Invention* - e.g., a new algorithm, new use of data, or program.
- *Innovation* - creative act and invention carried into wider use, leading to substantial kinds of change. Innovation implies the successful exploitation of new ideas.

# A combination of Activities

- *Innovation = invention + exploitation + diffusion.*
- *Invention*: The creative act or process and its result.
- *Exploitation*: Commercial development and adaptation to practical situations.
- *Diffusion*: Adoption by a wider audience.

Time lag

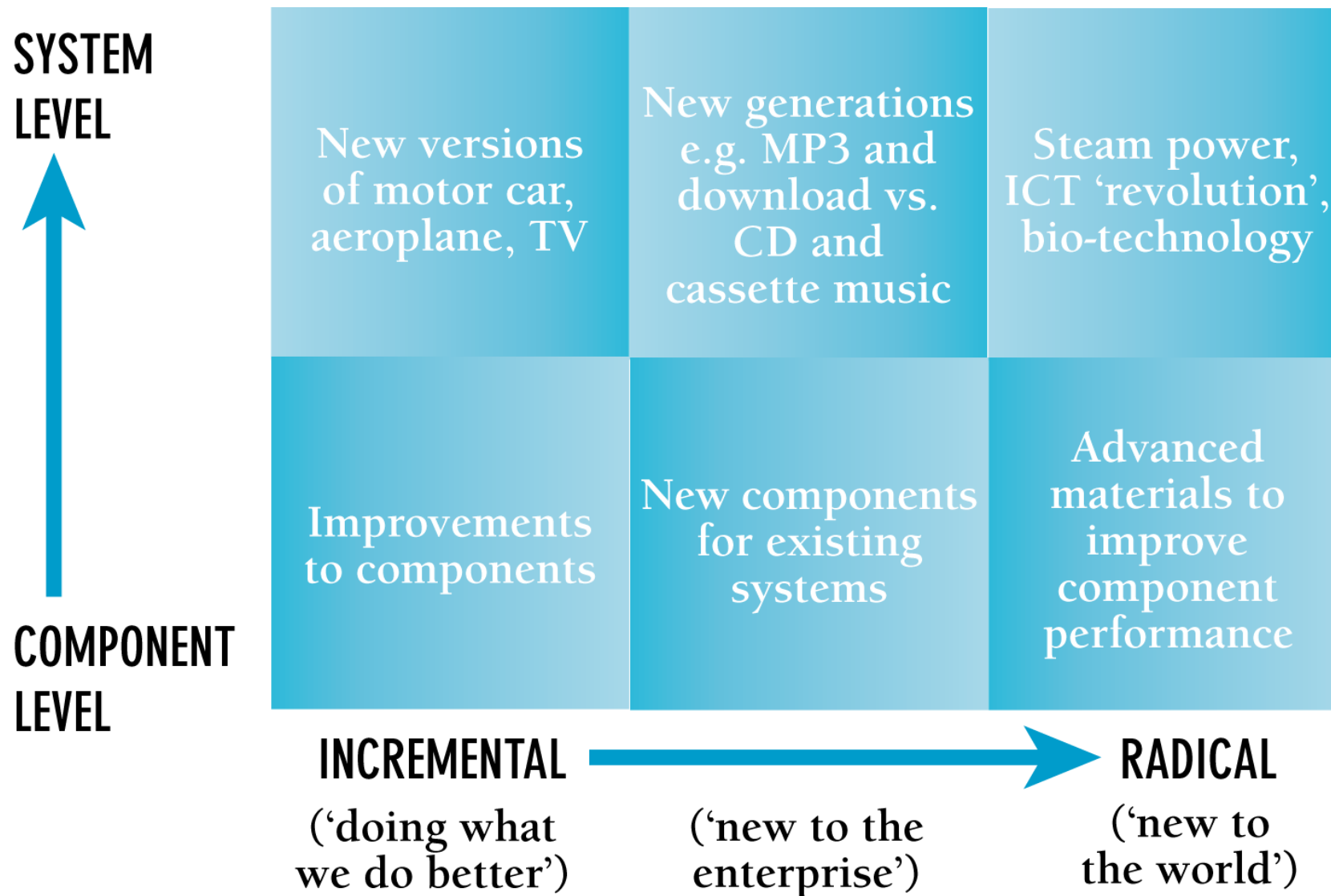


# Push and Pull

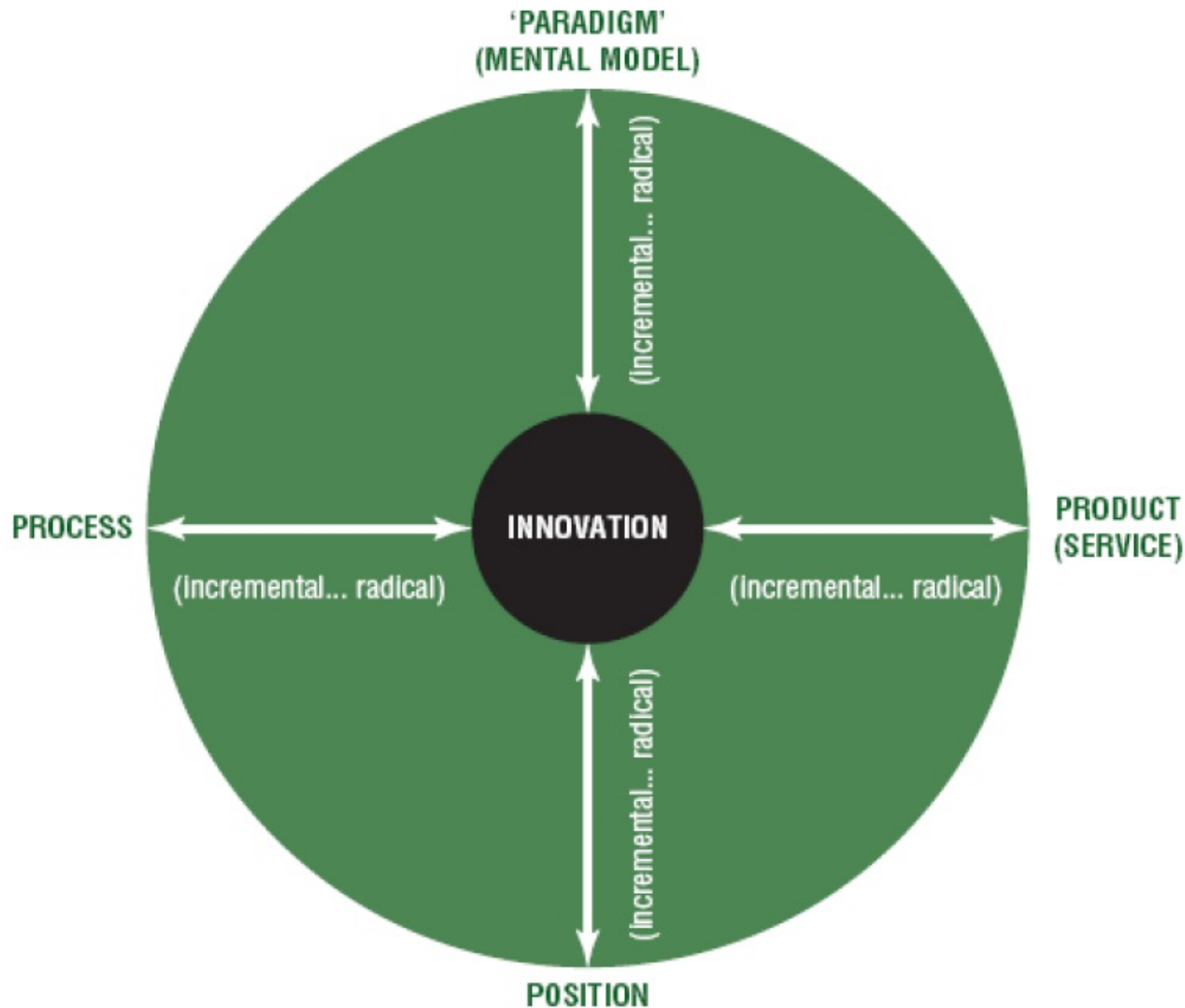
We may coarsely divide factors of economic or societal change into technology-push and demand-pull:

- *Technology-push*  
Changes linked to the emergence and diffusion of new technologies: Digitalization, Artificial Intelligence, Internet of Things, Industry 4.0, smart sensors, etc.
- *Demand-pull*  
When new solutions answer problems or needs: The seventeen UN Sustainable Development Goals (SDGs), climate change, pandemics, globalization, etc.

# Dimensions of Innovation



# Innovation Space



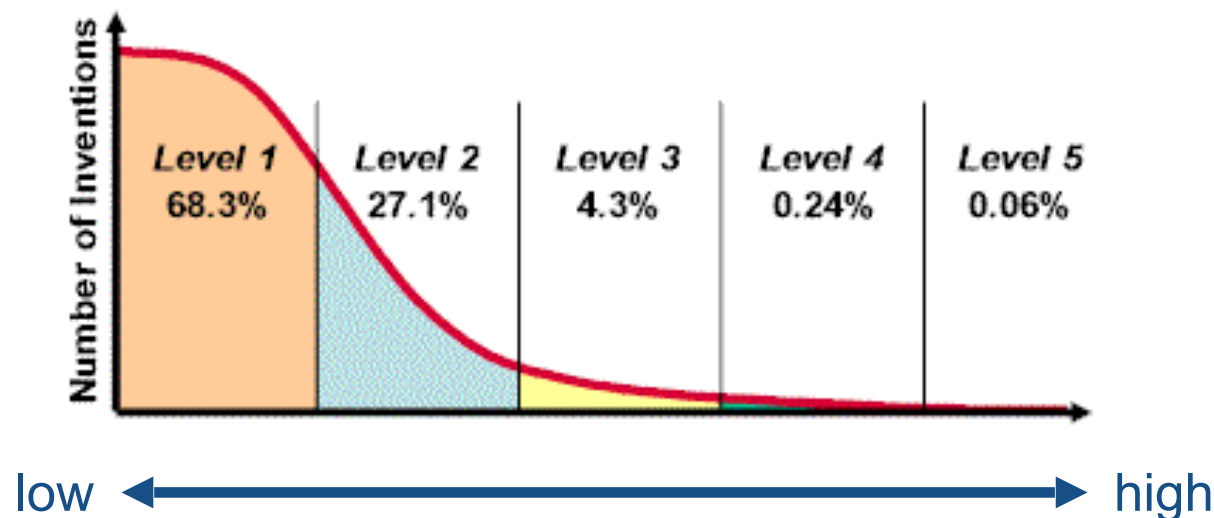
# Hierarchies of Technical Systems (Altshuller)



|         | system                    | subsystem                                                                 |
|---------|---------------------------|---------------------------------------------------------------------------|
| broad   | <b>transportation</b>     | <b><i>cars</i></b> , roads, maps, drivers, service stations               |
|         | <b>cars</b>               | power train, <b><i>brakes</i></b> , heating, steering, electrical         |
|         | <b>brakes</b>             | brake pedal, hydraulic cylinders, fluid, <b><i>brake pad assembly</i></b> |
|         | <b>brake pad assembly</b> | <b><i>pad</i></b> , mounting plate, rivets                                |
|         | <b>pad</b>                | particles a, b, <b><i>chemical bond</i></b>                               |
| focused | <b>chemical bond</b>      | molecules a, b                                                            |

# Novelty: Levels of Innovation (Altshuller)

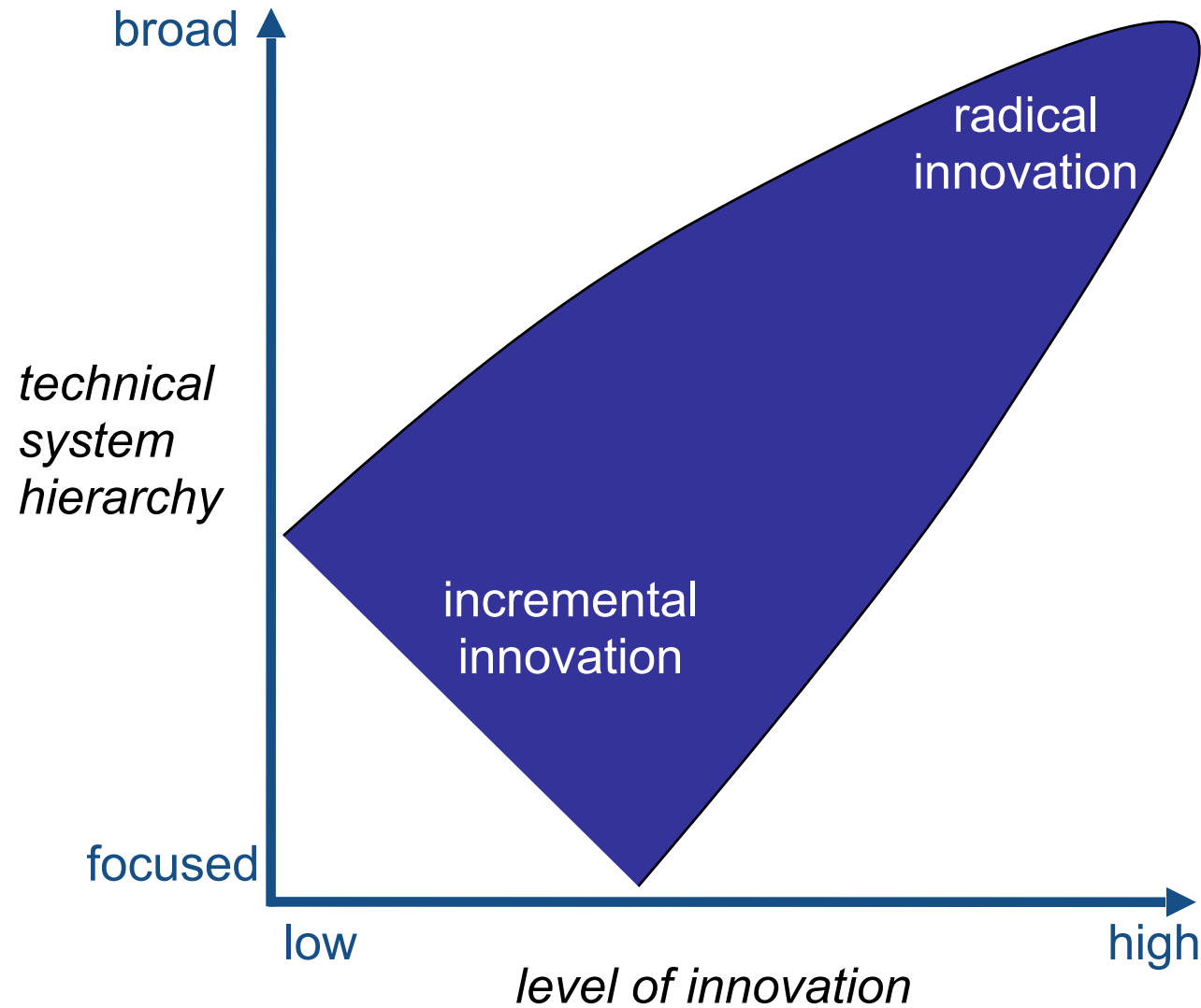
- level 1** - routine design problems solved by methods well known within the specialty - usually no invention needed.
- level 2** - minor improvements to an existing system using methods known within the industry.
- level 3** - fundamental improvement to an existing system using methods known outside the industry.
- level 4** - a new generation of a system that entails a new principle for performing the system's primary functions - solutions are found more often in science than technology.
- level 5** - a rare scientific discovery or pioneering invention of an essentially new system



Genrikh Saulovich Altshuller  
Born in 1926 and died in 1998  
Imprisoned 1950 - 1954



# Incremental and Radical Innovation



# Technology Development and Economic Progress

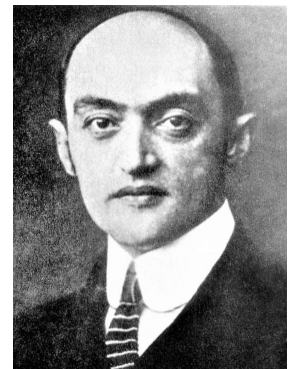
- Technology innovation is a good indicator for economic growth
  - Society
  - Industry
- Developed (= rich) societies/companies are
  - More dependent on innovation
  - Better innovators
  - Early users of innovations
- Non-innovative work often performed more cheaply in less developed countries (e.g. outsourcing)
- Incremental innovation produces change
- Radical innovation produces transformation

# Technology and Economics

- Neoclassical Economics:
  - Technology as **manna from heaven**
- Evolutionary Economics:
  - **Long Waves** (Kondratiev)
  - **Creative Destruction** (Schumpeter)



Nikolai Kondratiev  
Born in 1892, executed in 1938



Joseph Schumpeter  
1883 –1950

# Long Waves and Creative Destruction

- Long waves postulated by Kondratiev in the 1920s.
  - Steam, steel/railways, electricity, ...
- Creative Destruction (Schumpeter 1938)
  - The "process of industrial mutation that continuously revolutionizes the economic structure from within, incessantly destroying the old one, incessantly creating a new one."

# IT as 5th Kondratiev Wave

- **First four** waves (from Wikipedia):
  - Industrial Revolution (1771)
  - Age of Steam and Railways (1829)
  - Age of Steel and Heavy Engineering (1875)
  - Age of Oil, Electricity, the Automobile, and Mass Production (1908)
- **The age of IT** (1971) is described as the **5th** Kondratiev Wave.
  - Some researchers suggested that IT will die out like earlier waves.
  - BUT: IoT, Big Data, digitalization, AI, etc., seem to renew the IT wave.

# (High-tech) Innovation: The Motor of the Economy (Schumpeter)

- Economic **equilibrium** is always **disturbed** by the entrepreneur
  - New products
  - New methods of production
  - New sources of supply
  - New markets
  - New ways to organize business
- Meets **resistance** from established players
- Eventually competitors **catch up** and create a new equilibrium

2026: Maybe a new equilibrium will not be here anytime soon



**ORACLE®**

The average life expectancy of a multinational corporation - Fortune 500 or its equivalent - is between 40 and 50 years (Businessweek).

# Schumpeter's distinction between *invention* and *innovation*

- An *invention* is an idea, a sketch or model for a new or improved device, product, process or system. Such inventions may often (not always) be patented but they do not necessarily lead to technical innovations. In fact, the majority do not.
- An *innovation* in the economic sense is accomplished only with the *first commercial transaction* involving the new product, process, system or device, although the word is used also to describe the whole process.

# (Software) Innovation: Consequences

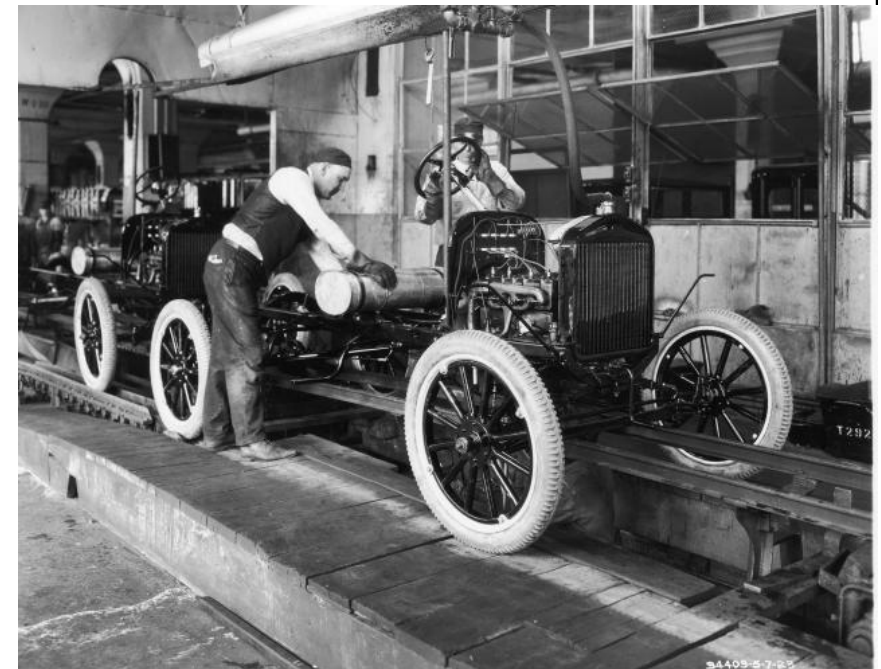
- **Installed base** = starting point.
- Result of software innovation is experienced as **change** in
  - The way people work.
  - The way business is carried out.
  - People's choice of entertainment.
  - Communication habits and interaction.
  - Governance of communities.
- Types of (software) innovation
  - **Radical** (disruptive, discontinuous).
  - **Incremental**.
  - **Resistance**.





# (Software) Innovation: Product and Process

- (Software) **product innovation** – (the creation of novel and useful software programs)
- (Software) **process innovation** – (the introduction of novel and useful ways of developing software)
- **Process** innovation might **enable** new **product** types



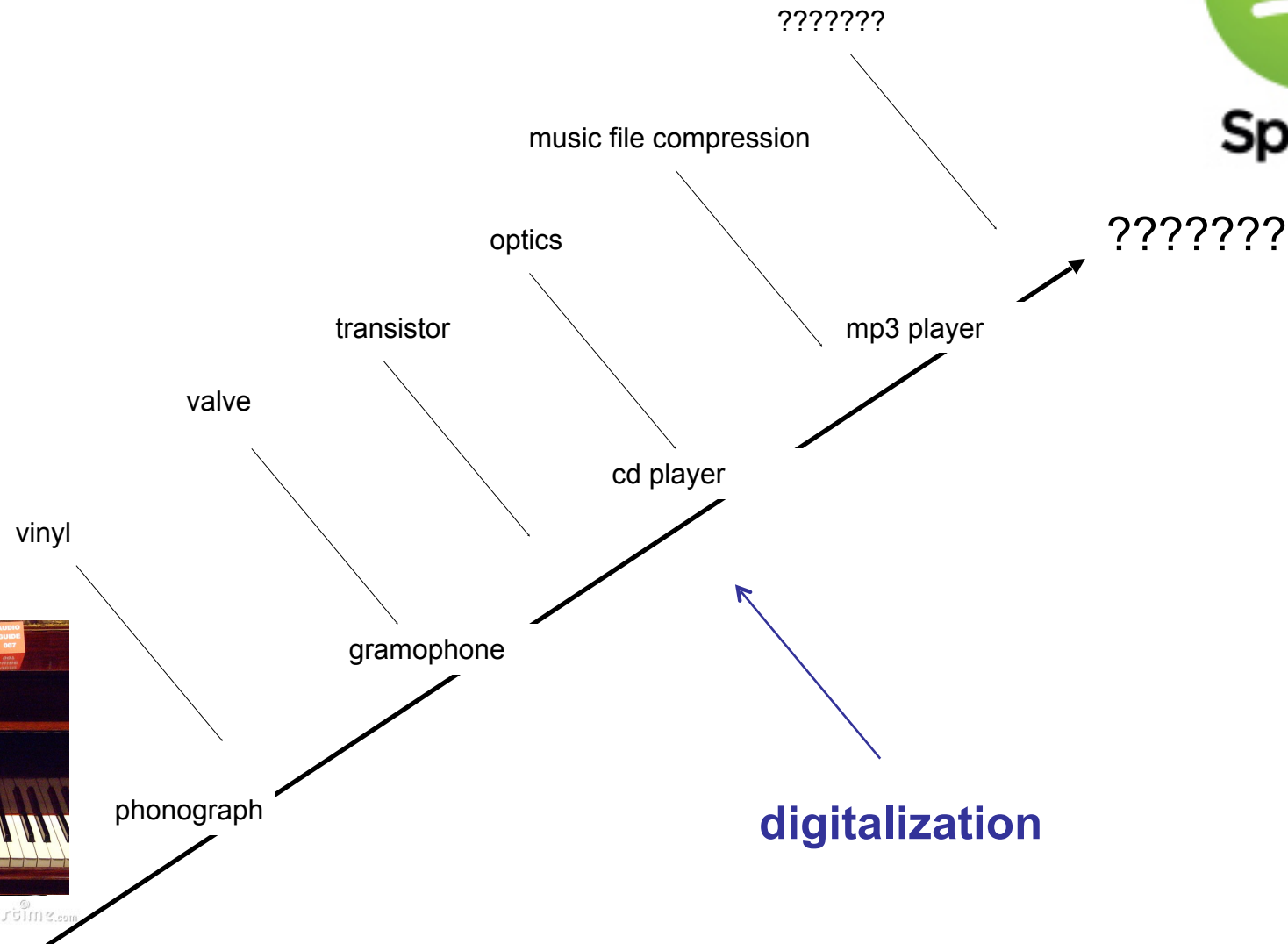
# (software) innovation timing

- Trajectory.
- Digitalization.
- Convergence.
- Infrastructure.
- Innovation windows.

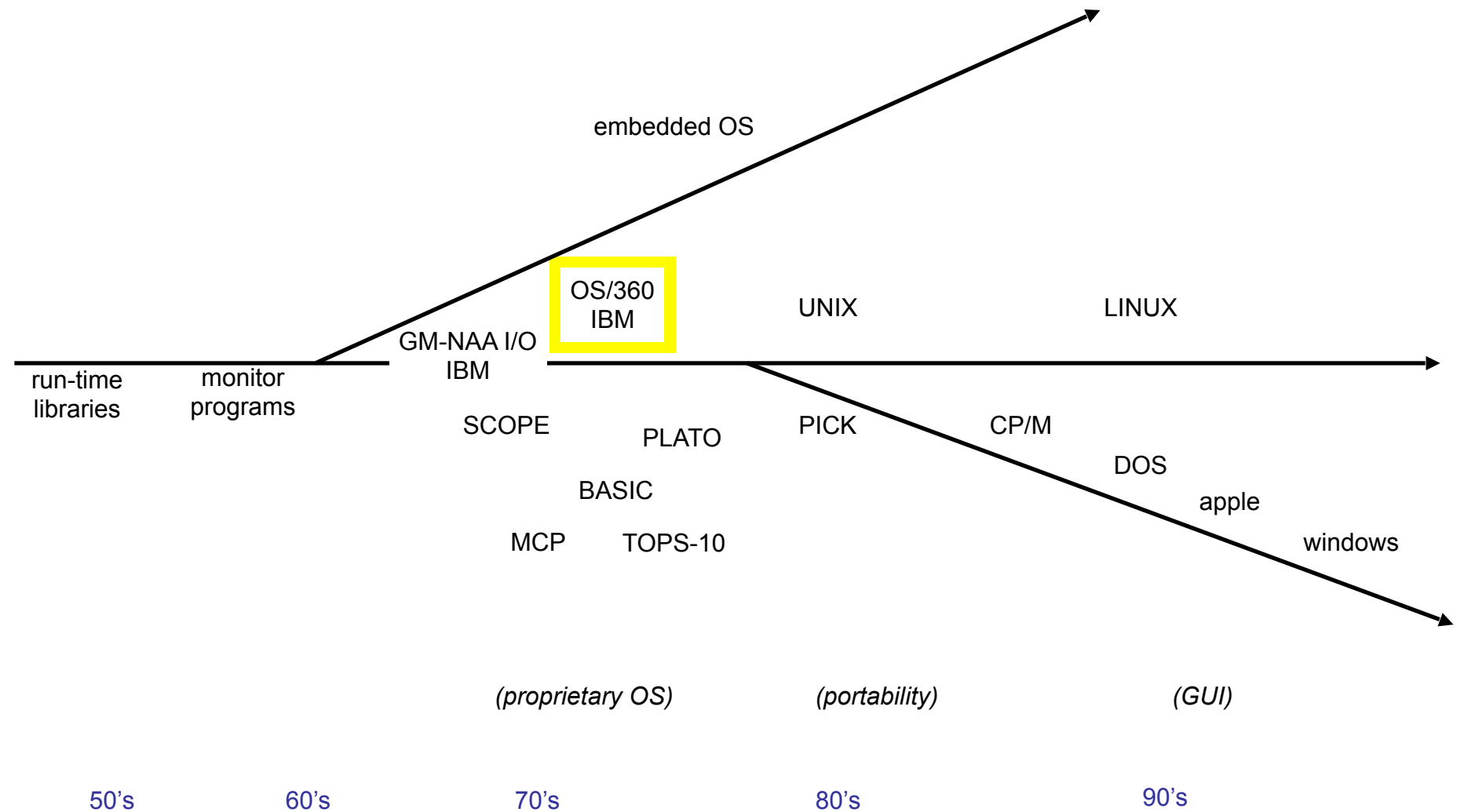
# Technology Trajectory



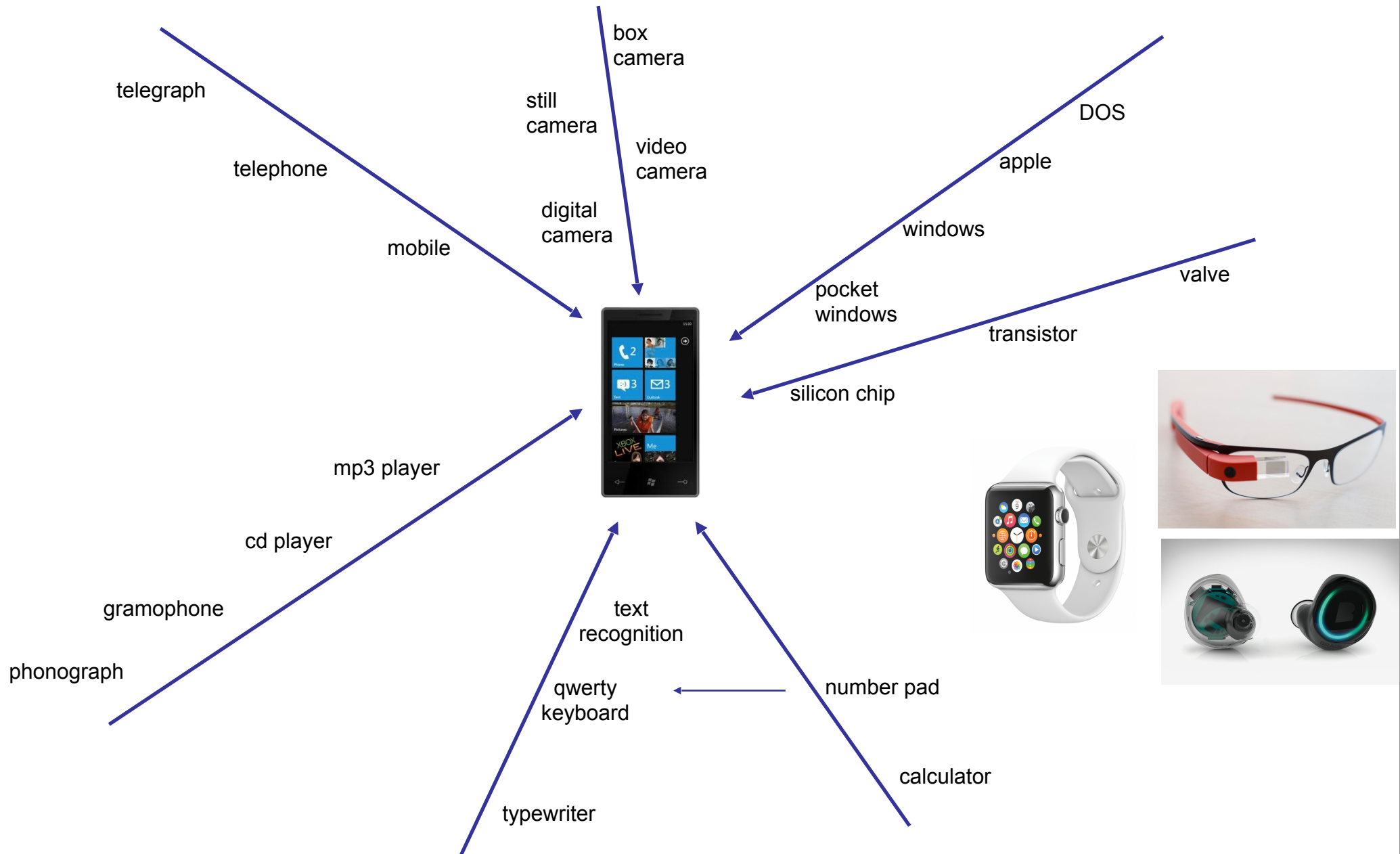
dreamstime.com



# Software Technology Trajectory

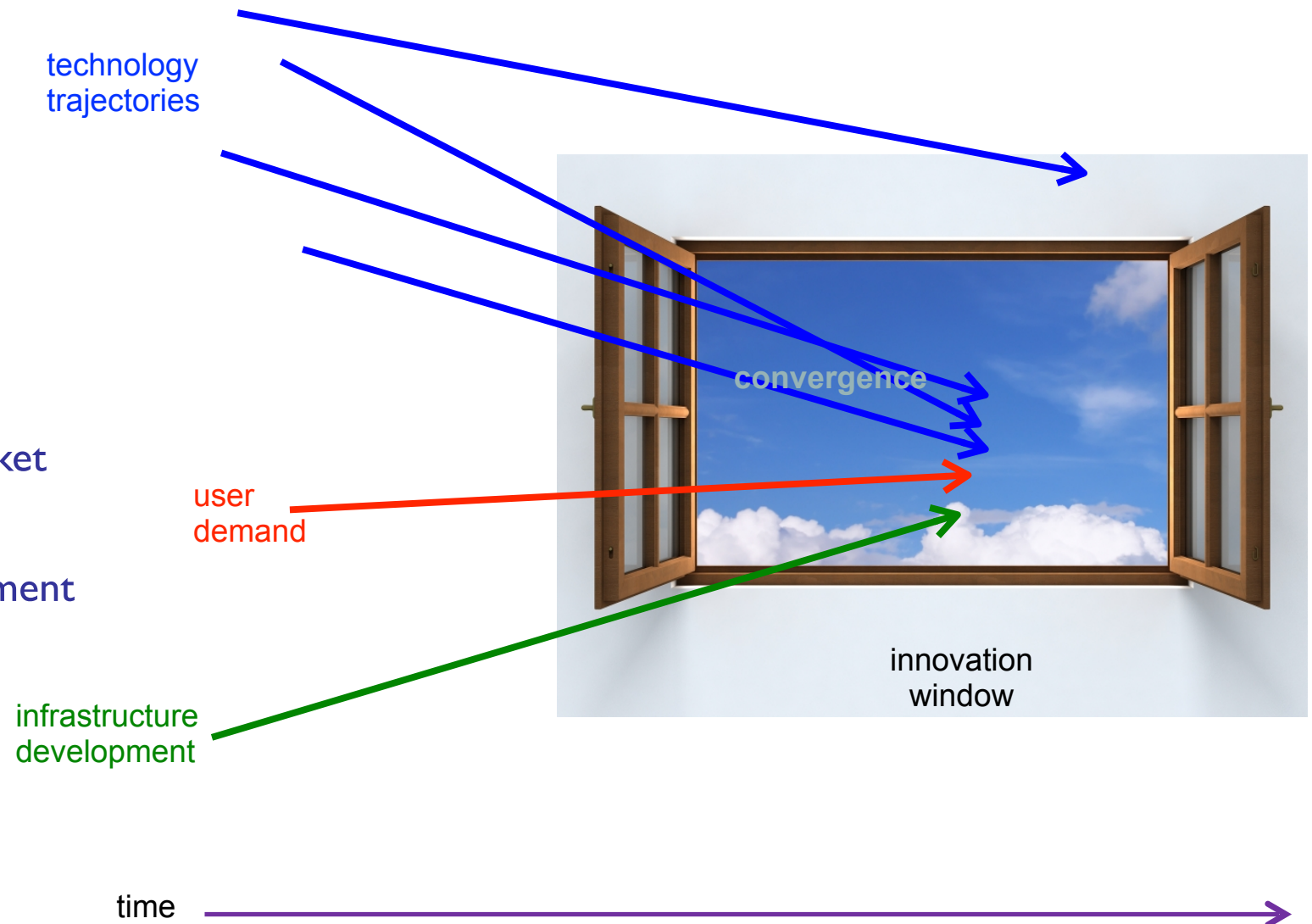


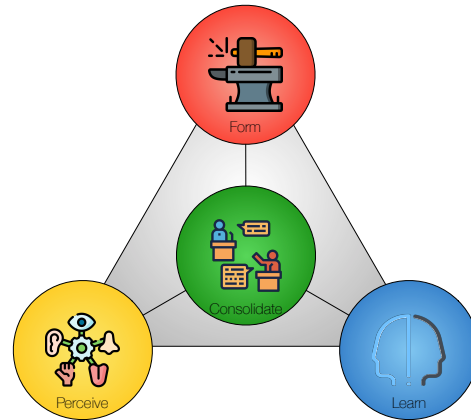
# Technology Convergence



# The Innovation Window

- Software technology trajectories
- Convergence
- (User) demand (= market development)
- Infrastructure development





# Assessment and Evaluation

# Several Types of Evaluation

- Personal and group creativity
- Software product assessment
- Work environment inventory



**1. I reach decisions more:**

systematically. ☐ ☐ ☐ ☐ ☐ ☐ by intuitive leaps.

**2. I learn best by:**

having clear goals and expectations of what I'll learn. ☐ ☐ ☐ ☐ ☐ ☐ having new experiences with no preplanned goals.

**3. I am most comfortable learning a new job when the learning primarily involves:**

skills for improving relationships. ☐ ☐ ☐ ☐ ☐ ☐ specific technical or academic information.

**4. When a job requires very detailed work, I tend to be:**

impatient and bored. ☐ ☐ ☐ ☐ ☐ ☐ patient and steady.

**5. I prefer working in groups that are:**

loosely organized. ☐ ☐ ☐ ☐ ☐ ☐ well-structured.

**6. I like solving problems best when:**

there are some standard ways to go about solving them. ☐ ☐ ☐ ☐ ☐ ☐ there needs to be a new way to go about solving them.

**7. Sometimes I might hinder constructive change by:**

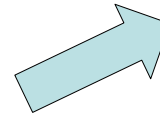
forgetting important details while focusing on future goals. ☐ ☐ ☐ ☐ ☐ ☐ getting lost in the details of implementation and forgetting the goal.

**8. I work best when:**

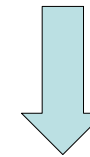
I am free to respond to unplanned situations. ☐ ☐ ☐ ☐ ☐ ☐ I plan my work ahead of time.

**9. I communicate better with people who think in:**

precise, no-nonsense ways. ☐ ☐ ☐ ☐ ☐ ☐ imaginative, unique ways.



|           |       |               |
|-----------|-------|---------------|
| facts     | _____ | insight       |
| decision  | _____ | perceiving    |
| modifying | _____ | exploring     |
| visioning | _____ | experimenting |

**innovation styles:**

- modifying
- exploring
- experimenting
- visioning

## personal creativity: psycho-metric testing

based on the innovation styles profile (ISP), <http://www.creativeadvantage.com/profile.htm>

# Innovative Software Product Assessment

|                                                                                             |                                         |           |               |
|---------------------------------------------------------------------------------------------|-----------------------------------------|-----------|---------------|
|                                                                                             | <b>The project idea presented is:</b>   |           |               |
| ICW                                                                                         | crude                                   | . . . . . | well-crafted  |
| IAI                                                                                         | appropriate                             | . . . . . | inappropriate |
| ICA                                                                                         | clear                                   | . . . . . | ambiguous     |
| IUO                                                                                         | unique                                  | . . . . . | ordinary      |
| ICE                                                                                         | coarse                                  | . . . . . | elegant       |
| IUU                                                                                         | usual                                   | . . . . . | unusual       |
| IAU                                                                                         | attractive                              | . . . . . | unattractive  |
| IOD                                                                                         | organized                               | . . . . . | disorganized  |
| IAC                                                                                         | astonishing                             | . . . . . | commonplace   |
| IMS                                                                                         | meticulous                              | . . . . . | sloppy        |
| <b>From the technical point of view, the proposed system, if implemented, will be:</b>      |                                         |           |               |
| TCS                                                                                         | complex                                 | . . . . . | simple        |
| TWU                                                                                         | workable                                | . . . . . | unworkable    |
| TMS                                                                                         | sensible                                | . . . . . | senseless     |
| TPU                                                                                         | pioneering                              | . . . . . | unprogressive |
| TAC                                                                                         | astounding                              | . . . . . | common        |
| <b>From the organizational point of view, the proposed system, if implemented, will be:</b> |                                         |           |               |
| ORA                                                                                         | revolutionary                           | . . . . . | average       |
| OIE                                                                                         | inessential                             | . . . . . | essential     |
| OOU                                                                                         | useful                                  | . . . . . | useless       |
| OKC                                                                                         | inexpensive                             | . . . . . | costly        |
| OOI                                                                                         | operable                                | . . . . . | inoperable    |
| OAC                                                                                         | astounding                              | . . . . . | common        |
| OCS                                                                                         | complex                                 | . . . . . | simple        |
|                                                                                             | <b>Overall, the proposed system is:</b> |           |               |
| SCU                                                                                         | creative                                | . . . . . | uncreative    |

(Lobert, B. M. and D. G. Dologite (1994). Measuring creativity of information system ideas: an exploratory investigation.)

# Innovative Software **Product** Assessment

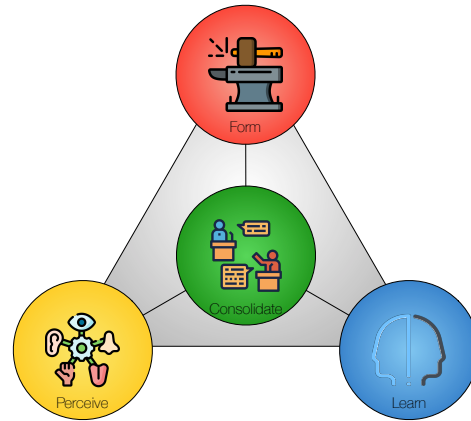
| <b>EVALUATION►<br/>PERSPECTIVE<br/>CREATIVITY<br/>DIMENSION▼</b> | <b>PROJECT IDEA</b>                                                       | <b>ORGANIZATIONAL</b>             | <b>TECHNICAL</b>         |
|------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------|--------------------------|
| <b>NOVELTY</b>                                                   | unique<br>unusual<br>astonishing                                          | revolutionary<br>astounding       | astounding<br>pioneering |
| <b>RESOLUTION</b>                                                | appropriate                                                               | operable<br>inexpensive<br>useful | workable<br>sensible     |
| <b>SYNTHESIS &amp;<br/>ELABORATION</b>                           | well-crafted<br>clear<br>elegant<br>attractive<br>organized<br>meticulous | complex                           | complex                  |

(Lobert, B. M. and D. G. Dologite (1994). Measuring creativity of information system ideas: an exploratory investigation.)

# Work Environment Inventory (WEI)

Six parameters:

- Freedom
- Challenging work
- Sufficient resources
- Supervisory encouragement
- Work group support
- Organizational encouragement



# Next Time

# Next Exercise and Lecture

- Exercise 9:
  - Evolvability & General Innovation Theory.
- Lecture 10:
  - Chapter 15 in Essence: Tactics (slides only).
  - Chapter 16 in Essence: Manifestations (slides only).
  - Chapter 15 in Essence: Capabilities (slides only).
  - Guest lecture: Klaus Bundvig, Netcompany.

# Exercise 9 (Evolvability & General Innovation Theory)

This exercise concerns the **evolvability** and **innovation characteristics** of your project.

- Describe and illustrate **diffusibility** in relation to your project.
- Describe and illustrate **adoptability** in relation to your project.
- Evaluate your project using one or more of the **evaluation types** presented on the slides.
- Describe your project using the terms **invention**, **innovation**, **exploitation**, and **diffusion**.
- Describe the **novelty**, **radicality**, and **timing** of your project.